

To intervene or not to intervene

Anonymous ACL submission

Abstract

This document is a supplement to the general instructions for *ACL authors. It contains instructions for using the L^AT_EX style files for ACL conferences. The document itself conforms to its own specifications, and is therefore an example of what your manuscript should look like. These instructions should be used both for papers submitted for review and for final versions of accepted papers.

1 Introduction

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2 Related Work

2.1 Turn taking

Facilitation interventions can be seen as a specific case of turn-taking, the process with which humans take turns speaking in a discussion. The goal of turn-taking is typically to minimize idle time in discussions, while letting most participants take part in the discussion. This process is opaque and largely learned after years of social interaction for humans, while only experimental coarse-grained solutions have been devised for automated systems (Nonomura and Mori, 2025). One solution is to use a dedicated smaller model whose only task is to decide whether a larger model should speak (Chen et al., 2025).¹ No such approach to our knowledge has been used in facilitative settings.

2.2 Facilitation detection

Deciding when to intervene in a discussion is a matter of great importance. Intervening too early or too frequently may lead to users expressing frustration at being constantly interrupted, while intervening too late may lead to a significant escalation and

derailment of the discussion (Korre et al., 2025; Gao et al., 2025). However, this task remains underexplored in literature, with current studies using ground-truth labels on which users are facilitators (Chen et al., 2025; Schroeder et al., 2024; Gao et al., 2025). The task proves harder when considering that a facilitative comment is not necessarily made by dedicated facilitators; since in online environments, facilitative feedback may be primarily provided by the users themselves (Falk et al., 2024).

While these works use LLM-as-a-judge models to annotate data in the field of facilitation, this may not be an obvious choice for the task of facilitation prediction. It has been demonstrated that LLM facilitators can not decide when they should intervene irrespective of instructions and model family/size. Instead, since these models are trained as chatbot assistants, they constantly attempt to speak (Tsirmpas et al., 2025).

3 Dataset

3.1 Description

Korre et al. (2025) provide a list of datasets relating to discussion facilitation. We combine all of these datasets (Table 1), preprocess them (A.1) and standardize their data according to a simple, unified schema (A.2). Relevant statistics can be found in A.3.

We thus create a multi-domain dataset focused around facilitation, called the “Prosocial and Effective Facilitation in Konversations (PEFK)” dataset. Unlike its predecessors, PEFK is large in size and rich in comment-level information, allowing us to run thorough analysis between similar datasets. In order to help research in the field of facilitation, we openly release PEFK under a CC-BY-SA license.²

¹Curiously, the authors specifically use LLM finetuning for the smaller model, but provide no explanation for why a traditional transformer-based classifier wouldn’t be sufficient for the task.

²Since the Fora dataset (Schroeder et al., 2024) is not publicly available, under an agreement we the authors we include the hashes of each comment instead of the original text. Researchers should contact the authors of Fora for access, after which they can merge their data with our completed

Name	#comments	Domain	Format
WikiDisputes (2021)	96,320	Forum	Text
WikiTactics (2022)	3,850	Forum	Text
WikiConv (2018)	17,806,373	Forum	Text
Conversations Gone Awry / CMV II (2019)	40,607	Forum	Text
CeRI (2012)	3,700	Forum	Text
User Moderation (UMOD) (2024)	2,000	Forum	Text
Intelligence Squared 2 (IQ2) (2016)	34,245	Debate	Oral
Why How Who (WHoW) (2025)	25,542	Radio / TV	Oral
Fora (2024)	39,438	Deliberation	Oral

Table 1: The datasets used in PEFK. We include a large variety of domains, and two different sources; text-only datasets, and transcripts from oral discussions.

3.2 Text-based vs. transcribed datasets

As the last column of Table 1 would suggest, our data belong in two main sub-modalities; regular written text in the form of posts and comments, and oral transcripts. This reflects a rift in current facilitation research; while some previous work concerns itself with forums and online discussions, others are exclusively focused on live deliberation and debate. Even though we focus on the former, we believe that work from both sides contributes to our understanding of facilitation.

The presence of these two sub-modalities necessitates separate experiments for either of them, since they represent different kinds of data. Posts and comments for instance are self-contained, while transcripts regularly feature interruptions, participants talking in parallel, and linguistic features that are necessary for coordination. They may also feature fundamentally different kinds of facilitation interventions.

Katerina will add social science research on how oral and written speech differs. Thus, we run separate experiments for each of the two dataset types.

4 How to decide when to intervene?

4.1 Do not decide

Dimitris. What most platforms do is to use flagging and reports to decide when to intervene. Mention Grok as a facilitative agent that works by invocation only. Briefly talk about advantages and disadvantages of current approach.

dataset.

4.2 Use humans

4.2.1 Merits and disadvantages

Katerina. Humans may be the “best” classifier since they understand context and social behavior best, but they are not scalable (cite our survey). Dimitris can add cost calculations for a hypothetical scenario to drive point home.

4.2.2 Experimental setup

Katerina. Annotation study. We leave experimental details for the appendix, but mention expert annotators, basic procedure, data and setup here. Also talk about positive vs negative reinforcement.

4.2.3 Results

Katerina. Inter-annotator agreement and discussion.

4.3 Use LLMs

4.3.1 Can we detect facilitation interventions?

Dimitris. We use LLMs and transformer-based models to decide whether the person talking is a facilitator.

4.3.2 Can LLMs decide when to intervene?

Dimitris, Katerina. No. We show this with two experiments. One is agreement between humans and LLMs on the annotation task (strong baseline). The other is by predicting when a facilitator would speak (weak baseline).

5 Conclusion

Humans are good but expensive. Our research is good. We wasted a good amount of money on this, please accept this.

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A Dataset details

A.1 Preprocessing

General

- We exclude comments with no text.
- We exclude discussions with less than two distinct participants.
- We exclude discussions which are common between wiktactics and wikiconv as well as wikidisputes and wikiconv.
 - There may be duplicate discussions between wikidisputes and wiktactics, but we allow them since they feature complementary information.

Wikiconv

- The Wikiconv corpus does not contain information about which user is a moderator/facilitator. Therefore, all comments relating to Wikiconv are tagged as non-moderators.
- Additionally, we follow the instructions of the original researchers (Hua et al., 2018), and select only discussions which have at least two comments by different users.
 - Wikipedia (thankfully) does not track users who log in with only an IP address (in the original dataset, their `user_id` is always set to 0 and their username is of the form 211.111.111.xxx. We consider each such username to be a separate user.
 - Due to the size of the dataset, we have to partially load it during preprocessing. Thus, there is a small chance every 100,000 records that a discussion is marked as a false negative and a part of it gets discarded.
 - We only include English comments in the final dataset. We use a small, efficient library (py3langid) for language recognition, due to the large size of Wikiconv. Non-English comments are discarded before selecting valid discussions (see point above).

Wiktactics

- We infer facilitative actions by whether the comment belongs in any of the following categories:
 - Asking questions
 - Coordinating edits
 - Providing clarification

- Suggesting a compromise
- Contextualisation

- The above tactics are a subset of the Coordinative labels used in the WikiTactics paper. They were selected because they are not used necessarily on 1-1 discussions; they could reasonably be applied by third-party participants. Contrast them with other Coordinative labels such as “Conceding/recanting” and “I don’t know”.

Wikidisputes

- Since only 0.03% of the comments in the dataset are made by moderators, we mark the dataset as not supporting moderation.

UMOD

- Facilitative actions are marked as a gradient from 0 (no facilitation) to 1 (full facilitation). We adopt a threshold of 0.75 to consider an action as facilitative, with more than 50% annotator agreement (measured as entropy in the original dataset).

CMV-AWRY2

- We mark a discussion as escalated when the derailment value (from the official dataset) is in the 60th upper percentile.
- We remove deleted (“[deleted]”) comments.

A.2 Schema

Table 2 details the information available for each comment in the dataset. Some information, such as whether the person talking is a moderator / facilitator, is available only on some sub-datasets. Also note that the `is_moderator` column may hold different definitions of what we consider as moderator / facilitator. UMOD for instance considers facilitative comments made by users, while others such as Fora consider a comment facilitative if it was made by a dedicated facilitator.

A.3 Statistics

Figures 1,2 show the number of comments in each discussion, and the number of words per comment in our dataset respectively. We can see that most discussions are comprised of only a few comments, although there is a significant tail of very long discussions. Similarly, most comments are relatively short (<100 words), although there is again a long tail of very long comments.

Figure 3 shows the percentage of comments made by facilitators for each of our datasets. In

Name	Type	Description
conv_id	string	The discussion’s ID. Comments under the same discussion refer to the same discussion ID.
message_id	string	The message’s (comment’s) unique ID.
reply_to	string	The ID of the comment which the current comment responds to. nan if the comment does not respond to another comment (e.g., it’s the Original Post (OP)).
user	string	Username or hash of the user that posted the comment
is_moderator	bool	Whether the user is a moderator/facilitator.
moderation_supported	bool	True if the moderation labels are directly computed from the original dataset
escalated	bool	A discussion-level measure denoting discussions which have been derailed
escalation_supported	bool	True if the escalation labels are directly computed from the original dataset
text	string	The contents of the comment
dataset	string	The dataset from which this comment originated from
notes	JSON	A dictionary holding notable dataset-specific information
toxicity	float	The “toxicity” score given to the comment by the Perspective API
severe_toxicity	float	The “severe toxicity” score given to the comment by the Perspective API

Table 2: The full set of columns used for each comment in the PEFK dataset.

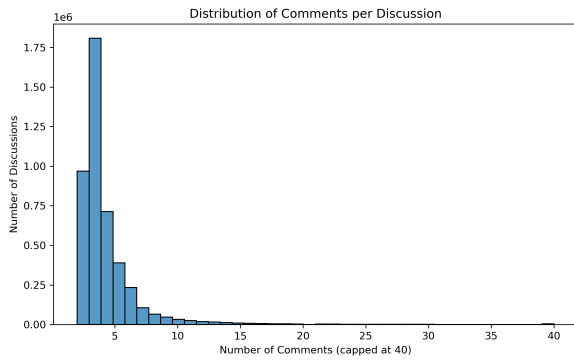


Figure 1: Comments per discussion on the PEFK dataset. Discussions over 40 comments have been clipped at the end of the x-axis for the purposes of graph readability.

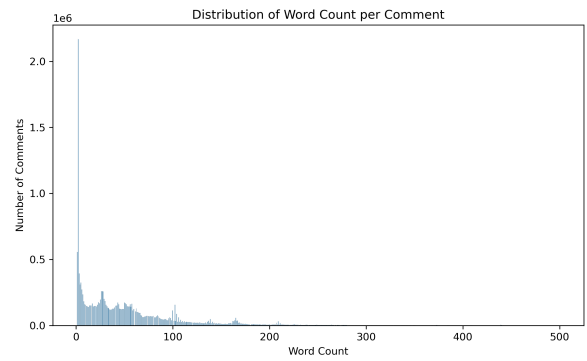


Figure 2: Words per comment on the PEFK dataset. Comments with 500 words have been clipped at the end of the x-axis for the purposes of graph readability.

almost all datasets, a third of the comments are made by facilitators. We do not include the UMOD dataset, since its percentage is based on our thresholding (see A.1).

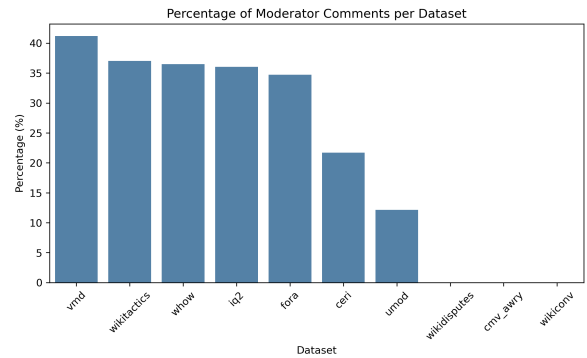


Figure 3: Percentage of moderated comments for each of the sub-datasets of PEFK.